

THE ADDRESSABILITY LIMIT: A GEOMETRIC CAPACITY BOUND FOR INFORMATION-GENERATING HIERARCHIES

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ABSTRACT. A process cannot retain distinguishable histories faster than its representation gains room to keep them apart. For retained-history growth rate β , radial conversion rate c , and host packing entropy h_{pack} , we prove $\beta \leq c h_{\text{pack}}$ unconditionally, in any proper metric space, with no curvature, tree, or stationarity hypothesis, and premises no stronger than retention, distinguishability at a fixed resolution, and a finite rate converting generative depth into representation radius. Read with its quantifiers rearranged, the same fact is a trichotomy: a process occupies exponential room, or its address radius never settles to a finite rate, or it forgets. The bound is *tight*: block address capacity equals $c h_{\text{pack}}$ *exactly* in every proper metric space, so packing entropy is a capacity, not a constraint. What packing entropy does not settle is structure. We exhibit a proper space of positive packing entropy admitting no uniform quasi-isometric image of a regular branching tree—room without shape—so the invariant that answers the addressing question is silent on the genealogical one. For one host family we close that gap: in real hyperbolic space, relational capacity attains block capacity at exponential order, $C_{\text{rel}}^{\text{wt}}(\mathbb{H}_{\kappa}^n, c; \varepsilon) = c(n-1)\sqrt{\kappa}$, so preserving a genealogy costs nothing beyond keeping its members apart. Under a full orthogonal symmetry hypothesis on the host’s boundary data, Schur’s lemma forces that family uniquely within Heintze’s classification. The curvature–entropy state equation is exactly the equality case of the bound in this isotropic host, asserted of no system here. Two questions are left open by name: for which hosts packing entropy is a complete invariant for relational capacity, and whether any physical process approaches the ceiling this bound proves exists.

On this manuscript. This paper supersedes a Zenodo record circulated as *A Geometric State Equation for Information-Generating Hierarchies* (concept DOI 10.5281/zenodo.19381557). That record’s public face was an *equality* between curvature and entropy rate; here that equality survives only as the named, detachable equality case of an inequality (§7), and the content is now the separation between two capacities rather than any single ceiling. A companion manuscript, *Evolution as Active*

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Geometry (Fenn & Fenn, in preparation), instantiates this paper’s bound and ladder on genomic data under its own evidentiary standard; it is cited, not fed back into any theorem below. No biological claim appears here.

1. INTRODUCTION, AND HOW TO READ THIS PAPER

A process that generates and retains distinguishable histories needs somewhere to put them. If a history is to stay distinguishable from every other retained one, it must occupy a location no other occupies at the working resolution; and a representation offers only finitely many such locations within a given reach. So unbounded memory inside a bounded radius is not merely hard but impossible—and the central fact of this paper is that the wall is a strict ceiling, standing exactly where the counting puts it.

What kind of statement this is. The result is a *limit principle*, in the lineage of the second law, the Carnot bound, and the Shannon capacity: an inequality that cannot be beaten, whose equality case is an idealized realization no particular system is assumed to reach. Most of this paper is the inequality and its exact tightness; a smaller part is the geometry of the equality case; and the honest premises separating the two are stated as they arise. The old “state equation” is the equality case, and it appears here in that role only.

The layered structure is the skeleton. The reader can hold the whole paper as four altitudes, and no claim below is stated above its altitude:

Layer	Statement	Status
I (§2–§3)	Bound $\beta \leq c h_{\text{pack}}$; exact block identity $C_{\text{block}} = c h_{\text{pack}}$; trichotomy; ladder; the balloon separation. Curvature-free.	Proven (Lean: finite/convergent case)
Bridge (§4)	Four-point condition $\Leftrightarrow$ tree; $n = 2$ embeddability floor. Neither Layer I nor II.	Proven (classical)
Host-specific (§5)	Relational capacity $C_{\text{rel}}^{\text{wt}}(\mathbb{H}_{\kappa}^n) = c(n-1)\sqrt{\kappa}$; zero tax. <i>Not</i> Layer-I universal.	Proven (uses [1])
IIa (§6)	A3 (full $O(d)$ symmetry) $\Rightarrow M = \mathbb{H}_{\kappa}^n$; the state equation as equality case.	Proven (Schur); A3 <i>asserted</i>
IIb (§7)	Variational selection of the boundary; $\eta = \beta/(c h_{\text{pack}})$. Does a process saturate?	Selection proven ; saturation open

Relation to the companion. A companion manuscript, *Evolution as Active Geometry*, imports exactly three objects from this paper and re-derives none: the *bound* of §2, the block/relational *ladder* of §2.3 and §5, and the *chart* statements of §4 and §6 (the minimal ambient dimension, and the isotropy convention under which a curvature is legally identified). Nothing in the companion’s empirical content is a premise below.

1.1. Relation to existing work. The pairing of trees with negatively curved hosts has a substantial literature; it is worth stating which questions it answers and which one it leaves.

Embedding results. Sarkar embeds any finite weighted tree in the hyperbolic plane with edge weights realized exactly and non-adjacent distortion below any $1 + \varepsilon$ [8]. Bonk–Schramm embed a Gromov hyperbolic space of bounded growth into $\mathbb{H}^n$ with additive distortion after rescaling [5]. Buyalo–Schroeder embed every Gromov hyperbolic group quasi-isometrically into a product of binary trees, with a sharp factor count [6, 7]. These settle *whether* an object embeds, at what distortion, into how many factors. Our question is different in kind: fixing a resolution ε and a radial budget $d(o, f(v)) \leq c\tau(v) + A_0$ turns embeddability into a *rate* problem—the supremum of exponential growth rates over admissible codes, not the distortion of any one. The two do not determine each other: low distortion for a fixed tree bounds neither, since a single embedding may consume arbitrary radius. We are not aware of a prior statement that this supremum equals ambient volume entropy exactly (Theorem 5.3), nor of the separation of §3 stated as a capacity rather than a dimension.

Critical exponents. The lower bound belongs to the Patterson–Sullivan tradition, in which a group’s critical exponent equals volume entropy and, for geometrically finite Kleinian groups, the limit-set Hausdorff dimension [3, 4, 2]. We require a sharper object: a free sub-semigroup with ε -separated orbit whose critical exponent approaches the ambient lattice’s, which is exactly Skenderi’s construction [1].

Terminology. *Capacity* always means an exponential growth rate of an admissible code, never the capacity dimension of [6]. *Host* means the ambient target. Dictionary for readers from metric geometry: block address capacity is the growth rate of ε -separated packings of balls; relational capacity restricts to codes whose image is a uniformly quasi-isometrically embedded weighted tree; the generative clock τ is the path-length metric of an edge-weighted tree.

2. THE LIMIT

2.1. Definitions.

Definition 2.1 (Retained-history growth). Let $\mathcal{H}_R$ be the histories retained through generative depth R , with $\mathcal{H}_R \subseteq \mathcal{H}_{R+1}$, and $N(R) := |\mathcal{H}_R|$. The retained-history growth rate is

$$\beta := \limsup_{R \rightarrow \infty} \frac{\log N(R)}{R} \quad (\text{nats per generative step}).$$

The word *retained* is load-bearing: a process that overwrites its past, so $\mathcal{H}_R$ does not grow, has $\beta = 0$ regardless of instantaneous change. That is the trichotomy’s third horn, not an exception to it. Define the effective bit rate $h_{\text{eff}} := \beta / \ln 2$, so $\beta = h_{\text{eff}} \ln 2$ *by definition*—a unit conversion, not a

physical claim. A Shannon entropy-rate estimator measures h_{eff} only when a typical-set or coding argument shows it tracks the exponential growth of retained distinguishable histories; alphabet entropy alone does not establish this. Locating the real empirical content here, rather than in the equality $\beta = h_{\text{eff}} \ln 2$, is what keeps the state equation of §6 honest.

Definition 2.2 (Faithful representation at finite radial rate). For a pointed metric space (M, d, o) , a family $f_R : \mathcal{H}_R \rightarrow M$ is ε -faithful at finite radial rate when (1) f_R is injective; (2) $d(f_R(x), f_R(y)) \geq \varepsilon$ for distinct x, y ; (3) $f_R(\mathcal{H}_R) \subseteq B(o, r(R))$; and (4) $c := \limsup_R r(R)/R < \infty$.

The units of c are geometric length per generative step; it converts the process clock to the host's radial ruler and cannot be omitted when curvatures under different radial gauges are compared. The convention $c = 1$ (the *process-time gauge*) is a coordinate choice, not a theorem.

Definition 2.3 (Packing entropy). With $P(B(o, \rho), \varepsilon)$ the supremum cardinality of an ε -separated subset of $B(o, \rho)$,

$$h_{\text{pack}}(M, o; \varepsilon) := \limsup_{\rho \rightarrow \infty} \frac{\log P(B(o, \rho), \varepsilon)}{\rho}.$$

This needs only a metric; it applies to trees, graphs, statistical manifolds, learned representations, and Riemannian manifolds alike. For a complete Riemannian manifold with uniform positive-lower and finite-upper volume bounds on balls of the relevant fixed radii, h_{pack} agrees with volume entropy h_{vol} at exponential order; we use them interchangeably only under that stated condition. The three quantities $\beta, c, h_{\text{pack}}$ are logically independent.

2.2. The bound and its converse.

Theorem 2.4 (Addressability bound). *If a process of rate β admits an ε -faithful representation at finite radial rate $0 < c < \infty$ in a host of finite packing entropy h_{pack} , with $r(R) \rightarrow \infty$, then $\beta \leq c h_{\text{pack}}$.*

Proof. Faithfulness gives $N(R) \leq P(B(o, r(R)), \varepsilon)$. For every $\delta > 0$, eventually $r(R) \leq (c + \delta)R$; monotonicity gives $P(B(o, r(R)), \varepsilon) \leq P(B(o, (c + \delta)R), \varepsilon)$. Finiteness of the limsup gives $\log P(B(o, \rho), \varepsilon) \leq (h_{\text{pack}} + \delta)\rho$ at large radii. Hence $\log N(R)/R \leq (c + \delta)(h_{\text{pack}} + \delta) + o(1)$; take the upper limit, then $\delta \downarrow 0$. $\square$

The finite-depth count and the convergent-rate case are machine-checked (`convergent_rate_addressability_limit`, Appendix B); the limsup statement is the paper theorem. The converse is not merely existence—it is an exact identity in full generality.

Definition 2.5 (Block address capacity). $A_{\text{block}}(\rho, \varepsilon) := \max\{|C| : C \subseteq B(o, \rho), d(x, y) \geq \varepsilon \forall x \neq y\}$ and $C_{\text{block}}(c, \varepsilon) := \limsup_R R^{-1} \log A_{\text{block}}(cR, \varepsilon)$.

Theorem 2.6 (Exact block-addressability identity). *For every proper metric space (M, d, o) , closed balls are compact, so $A_{\text{block}}(\rho, \varepsilon) = P(B(o, \rho), \varepsilon)$ for all ρ, ε , and consequently $C_{\text{block}}(c, \varepsilon) = c h_{\text{pack}}(M, o; \varepsilon)$.*

Proof. Properness gives compact balls, hence finite attained packing numbers; $A_{\text{block}} = P$ is the same extremal problem. Sampling the monotone packing-number function at linearly spaced radii $\rho = cR$ preserves its radial limsup. $\square$

Tightness is what separates a capacity from a constraint: nothing exceeds ch_{pack} and nothing prevents reaching it, so h_{pack} *measures* retained diversity rather than merely bounding it. This requires no tree, hyperbolicity, isotropy, or stationarity. The finite-radius existence half is machine-checked (`exists_optimal_blockCode` with `card_le_packingCount`, Appendix B).

Corollary 2.7 (Polynomial-growth exclusion; the trichotomy). *If $\beta > 0$ and $c < \infty$, then $h_{\text{pack}} > 0$: no host of only polynomial packing growth can faithfully represent retained exponential novelty at finite radial rate.*

This is the coordinate-free content: *remembering while creating requires an exponential-growth host*—on pain of the address radius failing to grow at any finite rate, or of forgetting. Positive packing entropy does *not* by itself imply negative curvature or Gromov hyperbolicity; hyperbolic space becomes canonical only after the symmetry assumption of §6.

2.3. The admissibility ladder. Theorem 2.6 proves achievability at exactly one level of task. Stronger senses of “represent a hierarchy” form a ladder, each class $\mathcal{A}$ inducing $C_{\mathcal{A}} \leq C_{\text{block}}$: **(1) Block**—terminal messages ε -separated; **(2) Persistent**—one nested address family retains prior messages; **(3) Causal**—parent-to-child addresses obey a per-step motion bound; **(4) Relational**—the representation preserves the source’s pairwise metric up to stated distortion (e.g. a uniform quasi-isometric embedding). Packing alone proves achievability only at level 1; §5 shows why the gap to level 4 is not automatically zero, and closes it for the canonical host.

3. THE SEPARATION: ROOM WITHOUT SHAPE

Packing entropy answers the addressing question completely and the structural one not at all.

Proposition 3.1 (The balloon). *There is a proper metric space (M, d, o) with packing entropy $h > 0$ at every resolution admitting no uniform quasi-isometric image of a regular branching tree.*

Construction. Take a ray of hub vertices at integer radii $0, 1, 2, \dots$ and attach to the hub at radius R a clique of $\lceil e^{hR} \rceil$ leaves at unit distance from it. With the graph metric this is proper, and each ball of radius ρ is dominated by the clique at radius $\lfloor \rho \rfloor$, giving packing entropy exactly h . But up to radius R there are only $O(R)$ bounded-diameter shells, while a regular branching tree has exponentially many mutually tree-distant vertices at depth R ; a quasi-isometric lower bound admits only a bounded number of tree-distant points per shell, so no uniform quasi-isometric embedding exists. $\square$

The balloon is exponentially room-rich and relationally empty: many addresses, no genealogy realizable at uniform distortion. It is the precise sense in which “diversity,” “packing,” and “preserved genealogy” are not synonyms, and the reason the ladder does not collapse to its top rung. Quartet defect δ (§4) is likewise not the capacity slack $\Delta := c h_{\text{pack}} - \beta$: a reticulate source can use every address, a perfect tree very few. (The balloon is also, verbatim, a star phylogeny after a rapid radiation; the companion develops that instance.)

4. THE STRUCTURAL BRIDGE: FOUR-POINT GEOMETRY

This section is neither Layer I nor Layer II. It decides whether a metric is tree-like—motivating the hyperbolic host class—without supplying β , c , saturation, or curvature magnitude.

For four points, form $S_1 = d(a, b) + d(c, d)$, $S_2 = d(a, c) + d(b, d)$, $S_3 = d(a, d) + d(b, c)$.

Theorem 4.1 (Four-point condition). *A finite metric is an additive weighted-tree metric exactly when the two largest of S_1, S_2, S_3 agree for every quartet [12]. In the geodesic setting, 0-hyperbolicity is exactly the $\mathbb{R}$ -tree condition [13].*

The defect δ measures departure from this. It determines whether the qualitative object is a tree; it does not determine a curvature magnitude.

Remark 4.2 (Minimal ambient dimension). A genuinely branching tree cannot embed isometrically in a one-dimensional connected Riemannian manifold: an interior path point admits two directions, a branch vertex needs three. So $n = 2$ is the minimal smooth hyperbolic ambient dimension admitting genuinely branching tree metrics at arbitrarily low distortion (path trees remain one-dimensional). This is a structural embeddability floor: *not* the minimizer of any curvature functional, not a fitted constant, and not to be inferred from a measured curvature via any state equation. The four-point condition alone does not imply $n = 2$.

5. CLOSING THE GAP: THE HYPERBOLIC HOST

This theorem is host-specific, not Layer-I universal. Hyperbolic space closes the gap the balloon opens, exactly.

Proposition 5.1 (Finite-depth obstruction). *Fix $\varepsilon > 0$ and $0 < c < \varepsilon$. If every rooted tree T must satisfy $f(T_R) \subseteq B(o, cR)$ for every integer depth R , with $f(\text{root}) = o$ and all distinct vertices ε -separated, the only admissible tree is the one-vertex tree.*

Proof. A child $v \in T_1$ needs $d(o, f(v)) \leq c < \varepsilon$; separation needs $d(o, f(v)) \geq \varepsilon$; contradiction. $\square$

This is not a failure of hyperbolic capacity but a mismatch between a physical rate, a fixed resolution, and a clock that charges every edge exactly one unit with no startup allowance—a constraint the addressability theorem never imposed. The former Conjecture 4.4 (exact unit-edge form) is thereby retired, and the correct formulation uses an independently calibrated clock.

Definition 5.2 (Weighted relational code). For a rooted, locally finite tree T with positive edge durations $a(e)$, $\tau(v) := \sum_{e \in [\text{root}, v]} a(e)$, $T_R^\tau := \{v : \tau(v) \leq R\}$, a map $f : V(T) \rightarrow \mathbb{H}_\kappa^n$ is a weighted relational code at rate c , resolution ε when $f(\text{root}) = o$; distinct vertices are ε -separated; $D^{-1}d_\tau(u, v) - K \leq d(f(u), f(v)) \leq Dd_\tau(u, v) + K$ for some $D \geq 1, K \geq 0$; and $d(o, f(v)) \leq c\tau(v) + A_0$ for depth-independent A_0 . Then $C_{\text{rel}}^{\text{wt}}(\kappa, n, c; \varepsilon) := \sup_{(T, a, f)} \limsup_R R^{-1} \log |T_R^\tau|$.

Theorem 5.3 (Relational capacity of real hyperbolic space). *For every $\kappa > 0$, $n \geq 2$, $c > 0$, $\varepsilon > 0$,*

$$C_{\text{rel}}^{\text{wt}}(\kappa, n, c; \varepsilon) = c(n-1)\sqrt{\kappa} = C_{\text{block}}(\mathbb{H}_\kappa^n, o; c, \varepsilon).$$

Proof. Write $h := (n-1)\sqrt{\kappa}$. Real hyperbolic space $\mathbb{H}_\kappa^n$ has bounded geometry (uniform two-sided ball-volume bounds, Bishop–Gromov), so its fixed-resolution packing entropy equals its volume entropy at every resolution, $h_{\text{pack}}(\mathbb{H}_\kappa^n, o; \varepsilon) = h_{\text{vol}}(\mathbb{H}_\kappa^n) = h$, and Theorem 2.6 gives $C_{\text{block}}(\mathbb{H}_\kappa^n, o; c, \varepsilon) = ch$. It therefore suffices to prove $C_{\text{rel}}^{\text{wt}} = ch$.

Upper bound. Let (T, a, f) be any weighted relational code. For $v \in T_R^\tau$ the radial condition gives $d(o, f(v)) \leq c\tau(v) + A_0 \leq cR + A_0$, so $f(T_R^\tau) \subseteq B(o, cR + A_0)$. Distinct vertices are ε -separated, so f restricted to T_R^τ is injective with ε -separated image, whence $|T_R^\tau| \leq P(B(o, cR + A_0), \varepsilon)$. Since $\log P(B(o, \rho), \varepsilon) = h\rho + o(\rho)$, the additive offset A_0 washes out of the exponential rate:

$$\limsup_{R \rightarrow \infty} \frac{\log |T_R^\tau|}{R} \leq \limsup_{R \rightarrow \infty} \frac{\log P(B(o, cR + A_0), \varepsilon)}{R} = ch.$$

As this holds for every admissible code, $C_{\text{rel}}^{\text{wt}} \leq ch$.

Lower bound. Fix a with $0 < a < h$; we construct an admissible code of weighted growth rate $\geq ca$. Because a is arbitrary, this gives $C_{\text{rel}}^{\text{wt}} \geq ch$ and, with the upper bound, equality.

(1) *A lattice, and its critical exponent.* Choose a torsion-free cocompact lattice $\Lambda < \text{Isom}(\mathbb{H}_\kappa^n)$ (these exist for every $n \geq 2$ [10]). Cocompactness gives orbit growth $\#\{\gamma : d(o, \gamma o) \leq \rho\} \asymp e^{h\rho}$, so the critical exponent—the abscissa of convergence of the Poincaré series $\sum_\gamma e^{-s d(o, \gamma o)}$ —equals the volume entropy $\delta(\Lambda) = h$ [3, 4].

(2) *An ε -separated finite-index subgroup.* The set $F_\varepsilon := \{\gamma \in \Lambda \setminus \{1\} : d(o, \gamma o) < \varepsilon\}$ is finite by proper discontinuity. As a finitely generated linear group, Λ is residually finite [11], so a finite-index normal $\Lambda_\varepsilon \triangleleft \Lambda$ meets F_ε trivially: for $\gamma \in \Lambda_\varepsilon \setminus \{1\}$, $d(o, \gamma o) \geq \varepsilon$, hence by invariance the whole orbit $\Lambda_\varepsilon o$ is ε -separated. A finite-index subgroup of a cocompact lattice is a cocompact lattice with the same exponent, so $\delta(\Lambda_\varepsilon) = h$.

(3) *A free sub-semigroup of nearly full exponent.* $\mathbb{H}_\kappa^n$ is Gromov hyperbolic and its Busemann coarse-cocycle is expanding with $\|\gamma\| = d(o, \gamma o)$ (Skenderi [1], Ex. 2.8), so his Theorem 3.1 applies to Λ_ε at any level $a < h = \delta(\Lambda_\varepsilon)$. It produces a finite $S \subset \Lambda_\varepsilon$ generating a *free* sub-semigroup $\Sigma = \langle S \rangle^+$ whose word length is comparable to displacement, $B^{-1}|w|_S - b \leq d(o, wo) \leq B|w|_S + b$; with the Schottky freeness this makes the orbit map of the regular $|S|$ -ary word tree a quasi-isometric embedding into $\mathbb{H}_\kappa^n$. Proposition 3.2(4), evaluated at the identity, is the partition inequality

$$\sum_{s \in S} e^{-a d(o, so)} \geq 1.$$

Since $\Sigma \subset \Lambda_\varepsilon$, its orbit is ε -separated, and freeness sends distinct words to distinct orbit points.

(4) *The weighted code, and its admissibility.* Let T be the regular word tree of Σ (vertices = finite words $w = s_1 \cdots s_k$ over S , root = empty word), and $f(w) := (s_1 \cdots s_k)o$. Give edge s the duration $a_s := d(o, so)/c > 0$; then $\tau(w) = \frac{1}{c} \sum_i d(o, s_i o)$, and only $|S|$ distinct durations occur. We verify the four code conditions. $f(\text{root}) = o$. Distinct vertices are ε -separated by step (3). The orbit map is a quasi-isometry for the unit word metric (step 3); as $\min_s a_s \cdot |w| \leq \tau(w) \leq \max_s a_s \cdot |w|$, the weighted metric d_τ is bi-Lipschitz to the word metric, so the map is a quasi-isometry for d_τ , with adjusted D, K . Finally, telescoping with isometry-invariance gives the radial budget *with zero offset*:

$$d(o, (s_1 \cdots s_k)o) \leq \sum_{i=1}^k d(s_1 \cdots s_{i-1}o, s_1 \cdots s_i o) = \sum_{i=1}^k d(o, s_i o) = c \tau(w),$$

so $A_0 = 0$. The code is admissible for the relational rung, its image being a uniformly quasi-isometrically embedded weighted tree.

(5) *The growth rate.* Because Σ is free, words biject with finite sequences over S , so the weighted counting series factors:

$$Z(q) := \sum_w e^{-q\tau(w)} = \sum_{k \geq 0} \left(\sum_{s \in S} e^{-qa_s} \right)^k,$$

which converges iff $g(q) := \sum_{s \in S} e^{-qa_s} < 1$. As g is continuous and strictly decreasing from $|S|$ to 0, there is a unique q^* with $g(q^*) = 1$, and q^* is the abscissa of convergence of Z . Since Z is the Laplace–Stieltjes transform of the nondecreasing counting function $R \mapsto |T_R^\tau|$, its abscissa equals that function’s exponential growth rate, $\limsup_R R^{-1} \log |T_R^\tau| = q^*$. Now evaluate at $q = ca$: since $qa_s = (q/c) d(o, so)$,

$$g(ca) = \sum_{s \in S} e^{-a d(o, so)} \geq 1,$$

so $ca \leq q^*$, i.e. this code has weighted growth rate $q^* \geq ca$. Hence $C_{\text{rel}}^{\text{wt}} \geq ca$ for every $a < h$, giving $C_{\text{rel}}^{\text{wt}} \geq ch$ and, with the upper bound, $C_{\text{rel}}^{\text{wt}} = ch = c(n-1)\sqrt{\kappa}$. $\square$

Remark 5.4 (The single modern input). Steps (1), (2), (5) are classical (Borel; Patterson–Sullivan; Malcev; the abscissa/growth-rate identity for a free weighted tree). The construction rests on exactly one contemporary result—Skenderi’s free sub-semigroup of nearly full critical exponent [1], used only through Theorem 3.1 (freeness and the quasi-isometric orbit tree) and Proposition 3.2(4) (the partition inequality). The zero-tax corollary inherits that single point of dependency, and no other part of this paper does.

Corollary 5.5 (Zero relational tax). *The relational tax $\Gamma := C_{\text{block}} - C_{\text{rel}}^{\text{wt}}$ is zero at exponential order in real hyperbolic space, as a supremum over weighted-clock, quasi-isometrically embedded tree codes. This establishes neither endpoint attainment, equal-edge achievability, nor zero tax for every prescribed genealogy, and must not be transferred to an arbitrary sequence metric or a host outside this construction.*

The supremum is essential and distinguishes this from the embedding literature (§1.1): a single low-distortion embedding may consume arbitrary radius; the rate question is created by the uniform radial budget along a family measured against one clock. The stronger equal-edge problem remains false for $c < \varepsilon$ (Proposition 5.1) and open for $c \geq \varepsilon$; it is a synchronization constraint, not host capacity.

5.1. Slack, efficiency, radial concentration. Write $B := c h_{\text{pack}}$ and the efficiency $\eta := \beta/B \leq 1$. Once a process is represented under the relational class, the block slack decomposes:

$$B - \beta = \underbrace{(B - C_{\text{rel}})}_{\text{relational tax } \Gamma \geq 0} + \underbrace{(C_{\text{rel}} - \beta)}_{\text{utilization slack } \Delta_{\text{use}} \geq 0}, \quad \eta = \underbrace{\frac{C_{\text{rel}}}{B}}_{\text{availability}} \cdot \underbrace{\frac{\beta}{C_{\text{rel}}}}_{\text{utilization}}.$$

In the weighted hyperbolic class $\Gamma = 0$ (Theorem 5.3), so block efficiency is exactly utilization; outside it, an observed $\eta < 1$ may mix tax with under-utilization, and the decomposition is the correct audit.

A converse locates where a near-capacity code puts its mass.

Theorem 5.6 (Radial concentration). *Let a weighted relational code in $\mathbb{H}_{\kappa}^n$, with $h = (n-1)\sqrt{\kappa}$, radial rate c , and offset A_0 , have growth at least $ch - \Delta$ in the finite form $|T_R^\tau| \geq B^{-1}e^{(ch-\Delta)R}$, where $\Delta \geq 0$ is an additive rate deficit. For $0 < \delta < 1$ put*

$$E_\delta(R) := \{v \in T_R^\tau : d(o, f(v)) \leq (1 - \delta)c\tau(v) + A_0\}.$$

Then, for a constant $C = C(\varepsilon, n, \kappa, A_0, B)$,

$$\frac{|E_\delta(R)|}{|T_R^\tau|} \leq C e^{-(h\delta c - \Delta)R},$$

whenever the growth lower bound holds—along a realizing sequence with any margin $\xi > 0$ for a limsup rate, and eventually if the limit exists.

Proof. The histories in $E_\delta(R)$ are ε -separated inside $B(o, (1-\delta)cR + A_0)$, so $|E_\delta(R)| \leq P(B(o, (1-\delta)cR + A_0), \varepsilon) \leq C_1 e^{h(1-\delta)cR}$ by the packing entropy of $\mathbb{H}_\kappa^n$. Dividing by $|T_R^\tau| \geq B^{-1} e^{(ch-\Delta)R}$ gives the stated ratio. No shell-regularity hypothesis is used. $\square$

At exact capacity ($\Delta = 0$) the fraction below any band vanishes, so $d(o, f(v))/(c\tau(v)) \rightarrow 1$ in probability under uniform counting on T_R^τ ; for fixed $\Delta > 0$, every band strictly below $1 - \Delta/(hc) = \eta$ empties. Thus “radius is accumulated information” is a consequence of a hyperbolic host *and* near-capacity operation—a statement about codes, conditional on both, and not an unconditional prediction about any physical system. The finite sub-ball count is machine-checked (`Packing.subball_fraction_le_packing_fraction`).

6. WHICH HOST

6.1. Isotropic realization. The packing entropy of §2 need not correspond to a curvature. We add a separate canonicity assumption.

Definition 6.1 (Isotropy convention, Assumption A3). The host M is complete, simply connected, unbounded, and homogeneous negatively curved—by Heintze [9] a solvable $N \rtimes_\varphi \mathbb{R}$ with N nilpotent and φ a positive-real-part derivation of $\mathfrak{n} := \text{Lie}(N)$. The generating process carries no directional structure beyond a real inner product on $\mathfrak{n}$: its symmetry acts as the full orthogonal group $O(d)$, $d := \dim \mathfrak{n}$, by Lie-algebra automorphisms.

Proposition 6.2 (Full symmetry forces isotropy). *Under A3, $\mathfrak{n}$ is abelian and φ a positive scalar multiple of the identity; equivalently $M = \mathbb{H}_\kappa^n$, with $h_{\text{vol}} = (n-1)\sqrt{\kappa}$.*

Proof. Two applications of Schur. The bracket $\Lambda^2 \mathfrak{n} \rightarrow \mathfrak{n}$ is $O(d)$ -equivariant. For $d \neq 3$, $\Lambda^2 \mathbb{R}^d$ and $\mathbb{R}^d$ differ in dimension, so $\text{Hom}_{O(d)} = 0$ and the bracket vanishes. At $d = 3$, $\Lambda^2 \mathbb{R}^3 \cong \mathbb{R}^3$ as $SO(3)$ -modules (cross product) but with opposite parity under $-\text{Id} \in O(3)$, so they are non-isomorphic as $O(3)$ -modules and the bracket again vanishes; $\mathfrak{n}$ is abelian. Then $\varphi \in \text{End}_{O(d)}(\mathbb{R}^d) = \mathbb{R} \cdot \text{Id}$ (the standard representation is absolutely irreducible for $d \geq 2$), and positivity forces a positive scalar. Abelian $\mathfrak{n}$ with scalar dilation is the horospherical presentation of $\mathbb{H}_\kappa^n$. $\square$

Remark 6.3 (What this replaces, and the diagnostic it licenses). This closes what an earlier draft carried as a conjecture—that hyperbolic geometry is the *generic* homogeneous negatively curved realization. Heintze’s classification does not support “generic”: the generic member has non-abelian N or a non-scalar contraction, hence pinched anisotropic curvature. Given A3 exactly, hyperbolic space is the *only* realization, and there is no genericity claim to make. A departure from A3 is therefore diagnostic, not merely disappointing: reducing the symmetry to a proper subgroup buys a specific structure and names which member of the rank-one class one is in— $U(n) \subset O(2n)$ the complex hyperbolic boundary, $\text{Sp}(n) \cdot \text{Sp}(1) \subset O(4n)$ the

quaternionic, $G_2 \subset SO(7)$ the octonionic. (At $d = 7$ the octonionic bracket is invariant only under $G_2 \subsetneq SO(7)$, already excluded by A3's full $O(d)$; and it is specifically the reflections in $O(3) \setminus SO(3)$ that kill the $d = 3$ cross product.) Which invariant tensor survives in a real system's boundary data is a sharper, falsifiable question than whether a fitted curvature looks anomalous. This paper states the diagnostic and does not run it; whether any real process satisfies A3 is an honesty item, not a mathematical gap.

Theorem 6.4 (Curvature floor and the state equation). *Under A3, Theorem 2.4 gives $\beta \leq c(n-1)\sqrt{\kappa}$, i.e. the curvature floor $\kappa \geq (\beta/(c(n-1)))^2$. The unit-invariant form uses $\bar{\kappa} := c^2\kappa$ (under $d \mapsto ad$: $c \mapsto ac$, $\kappa \mapsto \kappa/a^2$):*

$$\bar{\kappa} \geq \left(\frac{h_{\text{eff}} \ln 2}{n-1} \right)^2.$$

7. SATURATION: THE EQUALITY CASE, AND WHAT WOULD TEST IT

The state equation is the *equality case* of the bound in the isotropic host—no more, and asserted of no system here.

Proposition 7.1 (Exponential saturation). *If the limits defining β , c , h_{pack} along $r(R)$ exist and $\log P(B(o, r(R)), \varepsilon) - \log N(R) = o(R)$, then $\beta = ch_{\text{pack}}$. Combined with A3 this is the boxed equality*

$$\boxed{\bar{\kappa}^* = \left(\frac{h_{\text{eff}} \ln 2}{n-1} \right)^2} \quad \left(= (h_{\text{eff}} \ln 2)^2 \text{ at } n = 2 \right),$$

the “state equation” of the superseded record, retained here only as this conditional remark.

Why the boundary is special, without a dynamics. There is a principled, formula-independent reason a process might sit at the boundary: for *any* admissible-representation objective whose geometric cost is strictly increasing in an attainable host-capacity variable, subject to the lossless constraint $ch_{\text{pack}} \geq \beta$, the least-capacity minimizer (if attained) lies exactly on $h_{\text{pack}}^* = \beta/c$. Saturation is the natural answer to that one optimization problem. This does not depend on any particular description-length formula, and the saturation condition is analogous to a reversible Carnot cycle or a capacity-achieving Shannon code: the inequality is the limit, equality its ideal realization.

What is *not* proved. The mismatch $V(\kappa, \kappa^*) = (\sqrt{\kappa} - \sqrt{\kappa^*})^2$ is positive-definite with a unique zero at κ^* , but this is *not* Lyapunov stability, global attraction, or self-organization. Those require a differentiable potential on an invariant domain, an explicit evolution law $\dot{\kappa} = -\partial_{\kappa} U$, a proof that $\dot{V} < 0$ off equilibrium, and an empirical reason a modeled system follows it. Until such a dynamics is supplied, the correct claims are exactly four: uniqueness of the isotropic equality value; monotonicity of required capacity in rate; strict positivity of mismatch off equality; and variational selection of the

boundary under a stated capacity-cost objective. A mismatch potential is not a dynamics.

What would test saturation. Only $\eta := \beta/(c h_{\text{pack}})$ computed when β and h_{pack} carry *independent* provenance— β read from a process clock that is not the representation whose packing is measured. An η produced by fitting one axis to the other, or by back-solving n or κ from an assumed equality, tests nothing. Systematic, independently measured $\eta < 1$ in systems claimed to optimize description length, after finite-size, unit, distortion, and anisotropy control, would restrict the saturation hypothesis without touching the bound.

8. THE FIREWALL

Theorems 2.4, 2.6, 5.3 and Corollary 2.7 are unconditional. Theorem 6.4 is conditional only on Assumption A3. Nothing above requires that any physical process fills the capacity it proves exists, reports a curvature, back-solves a dimension, or treats an experiment as forcing hyperbolic realization over an alternative exponential host. We do not assert $h_{\text{vol}} = h_{\text{eff}} \ln 2$ as forced by the bound’s own axioms—that identity is a unit convention plus a definition (§2); we use no interior-entropy or unimodal-channel hypothesis; and we offer no second derivation from Fisher–Rao considerations—any resemblance between a location-scale geometry and §2’s radial construction is a remark, not a proof.

Relation to the companion. Evolution as Active Geometry needs exactly the three licenses of §1: the bound (room), the ladder’s block/relational distinction (names versus paths), and the chart’s minimal-dimension and isotropy statements (the identification a curvature analysis may legally make). It cites these, does not re-derive them, does not place a clade on a curvature scale using this paper’s mathematics, and does not run the matched-capacity comparison. Nothing in its empirical content is a premise here.

9. WHAT IS OPEN

Two questions, of different kinds. *Mathematical:* for which proper metric hosts is packing entropy a complete invariant for relational capacity? The balloon (§3) is the maximally negative answer, $\mathbb{H}_{\kappa}^n$ an affirmative one; everything between is open. The first tractable step is the rank-one symmetric family: complex and quaternionic hyperbolic spaces remain isotropic as Riemannian manifolds, so no sector-wise packing distinguishes them from real hyperbolic space of matching dimension, and whether their relational capacity also attains block capacity—and with which boundary entropy—is well posed with the machinery of §5 largely in place. Beyond the symmetric cases, a cocompact lattice need not exist, so discreteness surrogates may be required.

Empirical, and untouched here: the bound proves a ceiling exists and the identity proves it attainable; neither shows any physical process approaches

it. Whether retained diversity runs near ch_{pack} or orders of magnitude below is a measurement, requiring β and h_{pack} of independent provenance. We regard this as the more consequential question, the one the mathematics is meant to make askable, and one this paper does not answer.

APPENDIX A. MINIMAL STATEMENT

$$\boxed{\text{retained exponential novelty} \implies \beta \leq c h_{\text{pack}}}$$

unconditionally ($h_{\text{pack}} = h_{\text{vol}}$ under the stated volume-comparability). Separately,

$$\boxed{\text{A3 (full } O(d) \text{ symmetry, homogeneous negatively curved host)} \implies M = \mathbb{H}_{\kappa}^n}$$

by Schur (Proposition 6.2), with no genericity claim and no claim any process satisfies A3. Imposing in addition $\beta = c h_{\text{pack}}$ and $h_{\text{pack}} = (n-1)\sqrt{\kappa}$ gives $\bar{\kappa}^* = (h_{\text{eff}} \ln 2 / (n-1))^2$; and independently, the four-point condition $\Leftrightarrow$ tree metric. The quartet fixes the qualitative class; A3 fixes the host; the rate and radial calibration fix the scale; the inequality is the enduring limit; the boxed equality is its optimal isotropic saturated case, asserted of no system here.

APPENDIX B. FORMAL (LEAN) INVENTORY

Machine-checked, `theory/lean/ActiveGeometry/`, zero `sorry/admit`; block-identity proofs use only `propext`, `Classical.choice`, `Quot.sound`. The geometry lemmas of Theorem 5.3’s lower bound (Skenderi’s construction, residual finiteness, critical exponents of cocompact lattices) and the representation theory of Proposition 6.2 are classical, cited rather than formalized; the Lean predicate `IsotropicHyperbolic` names a hypothesis, not a derived conclusion.

File	Certifies
<code>Packing.lean</code>	Finite-depth count; <code>convergent_rate_addressability_limit</code> (Thm 2.4, convergent case); <code>exists_optimal_blockCode</code> , <code>card_le_packingCount</code> (Thm 2.6, finite radius); sub-ball fraction (radial concentration, finite).
<code>Capacity.lean</code>	Real algebra downstream of <code>Addressable</code> $\beta \leq c h_{\text{pack}}$: <code>efficiency_le_one</code> , <code>curvature_at_least_floor</code> , <code>saturated_curvature_eq_floor</code> , gauge invariance (<code>normalized_curvature_scale_invariant</code> , <code>process_time_gauge</code>). <code>CapacitySaturated</code> and <code>hcap_eq_spaceForm</code> are separate predicates.
<code>StateEquation.lean</code>	The equality-case face: <code>normalized_state_equation</code> (saturation $\wedge$ space-form $\Rightarrow c^2 \kappa = (h_{\text{eff}} \ln 2 / (n-1))^2$); <code>rateMismatchSq</code> diagnostics (positive-definite, unique zero) explicitly <i>not</i> a Lyapunov theorem.
<code>Measurability.lean</code>	Span-information identities behind the growth-class gate’s finite-sample refusal rule; an instrument property, not a theorem of §2–§7.

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